

Assignment 3

Remote grp

for September 26, 2026

Problem 0.1 (Munkres 16.9). Show that the dictionary order topology on the set $\mathbb{R} \times \mathbb{R}$ is the same as the product topology $\mathbb{R}_d \times \mathbb{R}$ where $\mathbb{R}_d$ denotes $\mathbb{R}$ in the discrete topology. Compare this topology with the standard topology on $\mathbb{R}^2$.

Proof. We recall that the dictionary order on $\mathbb{R}^2$, $\prec$, compares the first coordinate,

$$(x_1, x_2) \prec (y_1, y_2) \text{ if } x_1 < y_1,$$

and if $x_1 = y_1$, we compare the second coordinate, concluding $(x_1, x_2) \prec (y_1, y_2)$ if $x_2 < y_2$.

Then, the dictionary order topology is generated by open intervals $(a \times b, c \times d)$ where $(a, c) \prec (b, d)$. In particular the basiselements are

$$\{x_1\} \times (y_1, y_2)$$

and

$$\{x_1\} \times (y_1, \infty) \cup (x_1, x_2) \times \mathbb{R} \cup \{x_2\} \times (-\infty, y_2).$$

□